

Program: Diploma in Cyber Forensics and Information Security	
Course Code : 3288	Course Title: Data Structures using C Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Implement different Searching and Sorting Algorithms
- Implement linear data structures in C like Array, Stack, Queue, and Linked List
- Implement Non-linear Data Structures in C like Tree and Graph.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Control Structures, Modular Program Design, Arrays, Functions		Problem Solving and Programming	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the concept of Pointers, Structures and Files in C	9	Applying
CO2	Implement Searching and Sorting Algorithms in C	10	Applying
CO3	Develop Programs to Construct Stack, Queue, and Linked List	11	Applying
CO4	Develop programs to implement Graphs, Binary Search Tree and perform the basic on Graphs & Trees.	12	Applying
	Lab Exam	3	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3		3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the concept of Pointers, Structures and Files in C		
M1.01	Implement C program to demonstrate Pointers.	3	Applying
M1.02	Develop C program to demonstrate the concept of Files.	3	Applying
M1.03	Develop a C program to demonstrate the concept of Structures.	3	Applying
CO2	Implement Searching and Sorting Algorithms in C		
M2.01	Implement Linear Search and Binary Search in C	3	Applying
M2.02	Implement Bubble Sort, Selection Sort, Quick Sort and Merge Sort in C	7	Applying
	Lab Exam – I	1.5	
CO3	Develop C Programs to Construct Stack, Queue, and Linked List		
M3.01	Implement Stack and Queue using Array in C	2	Applying
M3.02	Develop C program for Infix to Postfix conversion and Postfix Evaluation using Stack	3	Applying
M3.03	Implement Linked List operations - insertion, deletion, searching, etc. in C	3	Applying

M3.04	Develop C program to construct Stack and Queue using Linked List.	3	Applying
CO4	Develop C programs to implement Graphs, Binary Search Tree and perform the basic operations and Traversals on Graphs and Trees.		
M4.01	Develop C Program to implement Graph	2	Applying
M4.02	Implement Graph Traversal Algorithms in C	3	Applying
M4.03	Develop C Program to Construct and Perform various Operations in Binary Search Tree	3	Applying
M4.04	Develop C program to perform the various traversals in a Binary Search Tree.	2	Applying
M4.05	Develop C Program to Print number of nodes and height of BST	2	Applying
	Lab Exam – II	1.5	

Text /Reference:

T/R	Book Title/Author
T1	Data Structures using C by E. Balaguruswamy - Tata McGraw-Hill Education
R1	Data Structures using C by Aaron M Tenenbaum - Pearson Education India
R2	Reema Thareja, “Data and File Structures using C”, Oxford University Press.
R3	Andrew S. Tanenbaum, “Data Structures using C & C++”, PHI.
R4	Jean-Paul Tremblay and Paul G. Sorenson, “An Introduction to Data Structures with Applications”, Tata McGraw Hill.

Online Resources:

Sl No	Website Link
1	https://www.cet.edu.in/noticefiles/280_DS%20Complete.pdf
2	http://www.tutorialspoint.com/cprogramming/
3	http://www.indiabix.com/online-test/c-programming-test